

23/2/26

Quiz - 3

STOP @ 9:25 AM

① 5 marks.

Consider the system

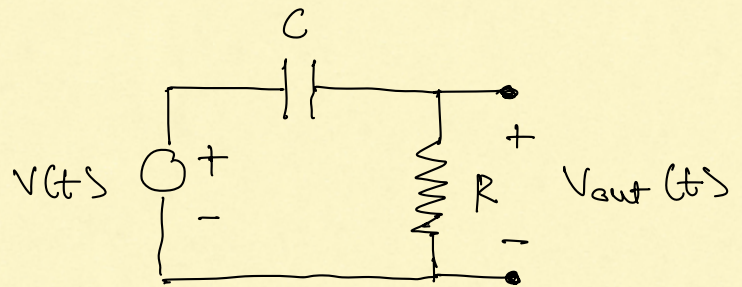
$$y(t) = 3 \cdot \operatorname{Re}(x(t)) + j \operatorname{Im}(x(t)).$$

Is this system linear? Explain why.

② 5 marks

$V(t)$ is T -periodic and differentiable, and

$V(t) \xleftrightarrow{\text{F.S.}} C_n$



Find the Fourier series coefficients of $V_{out}(t)$.

Instructions

Please write the last 3 digits of your roll number

prominently at the top and center of the

first page of answer sheet.

Examples:

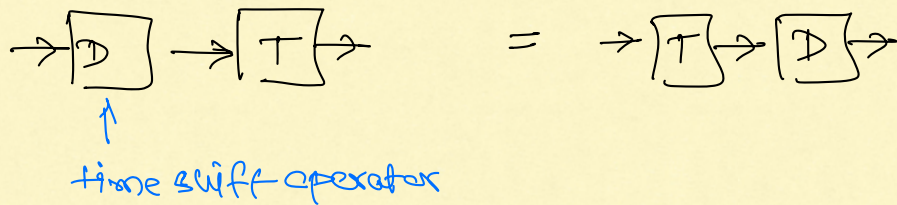
EE25BTECH11276 $\rightarrow$ 276

EE 25 BTECH11082 $\rightarrow$ 082

Lecture - 17

I. Time-Invariance

We say that a system is time invariant if a time-shift on the input is reflected as a time shift on the output.



Defn: For a given $t_0 \in \mathbb{R}$ let D denote the time-shift operator

$$x(t) \rightarrow \boxed{D} \rightarrow x(t - t_0).$$

We say that a system T is time invariant if T commutes with D , i.e. for any signal x we must have:

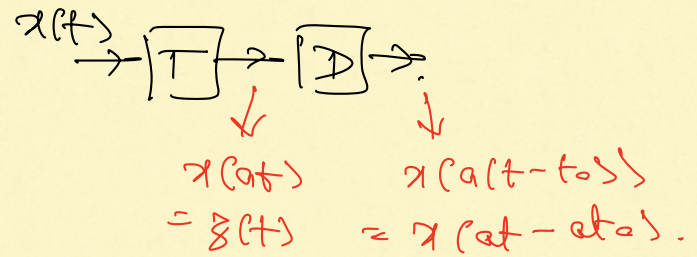
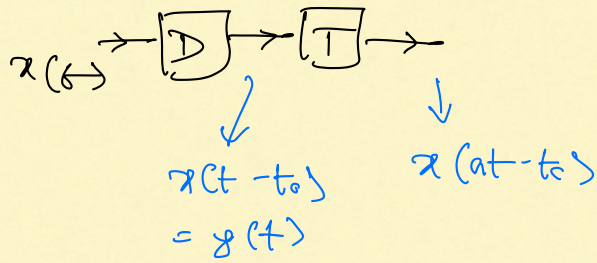
$$T(D(x)) = D(T(x))$$

Examples:

① T is the time-scaling system, i.e.

$$x(t) \rightarrow \boxed{T} \rightarrow x(at), \quad a > 0 \text{ is a constant.}$$

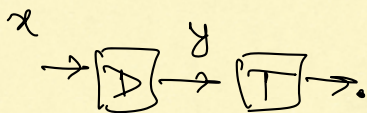
Checking for time-invariance:



$$y(t) \rightarrow [T] \rightarrow y(at):$$

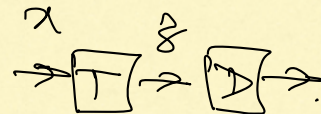
$$z(t) \rightarrow [D] \rightarrow z(t - t_0).$$

② $x(t) \rightarrow [T] \rightarrow t \cdot x(t).$



$$y(t) = x(t - t_0)$$

Output at time t
 $= t \cdot y(t)$
 $= t \cdot x(t - t_0).$



$$z(t) = t \cdot x(t)$$

Output at time t :
 $= z(t - t_0)$
 $= (t - t_0) \cdot x(t - t_0)$

Examples ① & ② are time variant systems

Since the outputs are not equal.